

Abdullah Korra

I am applying for a Master of Science in Mechanical Engineering because I want rigorous research training in the design, integration, and experimental development of mechanical systems that solve real problems in constrained environments. My long term goal is to build affordable machines for communities like the one where I grew up in The Gambia, where farming is labor intensive, resources are limited, and the available technologies are often poorly matched to local needs. I am especially interested in mechanical systems that combine design, actuation, sensing, controls, and prototyping to reduce physical strain and expand opportunity. Graduate study is the right next step for me because I have reached the point where curiosity and project experience are no longer enough on their own. I now want the depth, discipline, and mentorship of a research environment.

My interest in mechanical systems began well before college. I grew up in a farming village, and I became a car mechanic's apprentice at a young age. That experience shaped how I think. I learned to look at a machine not as a finished object, but as a system whose behavior could be understood, tested, and improved. I liked tracing faults, seeing how parts interacted, and learning how physical understanding could restore function. Later, when I encountered physics and mathematics in school, I found the intellectual foundation for the instincts I had already developed with my hands. Mechanical Engineering became attractive to me because it joined both sides of that interest. It offered a way to understand physical systems deeply while also designing new ones.

My path to higher education was not straightforward. In 2016, I left The Gambia in search of educational opportunities I could not realistically reach otherwise. The journey through Libya and across the Mediterranean was dangerous and life changing. I nearly died from hunger and harsh conditions, and I buried a friend along the way. By the time I reached Italy, education no longer felt routine. It had become something I had already paid too much to abandon. That experience did not make me interested in engineering, but it made me serious about it. It is one reason I care so much about doing work that is useful, rigorous, and worth dedicating years to.

At the College of Idaho, I pursued a BS in Math Physics with minors in Computer Science, History, and Art. That path gave me a strong quantitative foundation while also giving me room to build broadly. My technical preparation now includes embedded systems, controls, power electronics, CAD, 3D printing, fabrication tools, and programming with Python, Arduino, and Raspberry Pi. What matters most to me, however, is not just the list of tools. It is the way I have used them. I have consistently tried to move beyond isolated assignments and toward integrated systems that require design decisions, tradeoffs, testing, and revision. That is the kind of work I want to continue in graduate school.

The clearest example is my AgriBot project. I have been developing an autonomous farming robot intended for settings where labor is intense and cost matters. In this work, I engineered the motor subsystem for four independent drive wheels and a plow, wrote Arduino firmware, connected the system to a Raspberry Pi, built a Python GUI, and integrated GPS, IMU, and other components for navigation and operation. I also worked on path planning and system level communication. What makes this project important to me is not only that I built it, but that it forced me to think like a researcher rather than only like a builder. The hard questions were not simply how to make the robot move. They were how to design

a system that is robust enough for real use, simple enough to maintain, and affordable enough to matter. Working on AgriBot pushed me toward research questions about design under constraint, actuator and sensor integration, controls, and the gap between a working prototype and a dependable machine.

My research experience in astronomy further prepared me for graduate study. As an undergraduate astronomy research assistant, I processed and analyzed large datasets, helped develop a Python pipeline that produced science ready one dimensional and two dimensional spectra, and built a user facing interface for spectroscopy reduction. I also contributed to spectroscopic work that led to authorship on a wavelength calibration paper. Although this research is outside Mechanical Engineering, it taught me habits that are essential to good research in any field. I learned how to work through technical ambiguity, document methods clearly, debug workflows, and improve tools so that results are reliable and reproducible. Most importantly, I learned that I enjoy the research process itself. I like careful iteration. I like figuring out why a system behaves differently than expected. I like building methods that other people can actually use.

I have also sought out smaller experimental projects that reflect the same inclination. I designed and assembled an LED based photoelectric effect apparatus and built a functional ionic thruster. These projects strengthened my interest in experimental work because they required the connection of theory, apparatus, and testing. They reminded me that I am most engaged when I can move back and forth between analysis and physical implementation. That is a large part of why I am drawn to research focused MS programs rather than course based ones. I do not want only advanced classes. I want sustained involvement in a lab where I can contribute to the design, testing, and refinement of real systems.

My research interests within Mechanical Engineering center on mechatronic and robotic systems, especially for agricultural and other resource constrained settings. I am interested in the design of machines that are not only functional, but durable, repairable, and appropriately scaled to the environments in which they will be used. I want to deepen my knowledge of dynamics, controls, actuation, sensing, design for manufacturability, and experimental methods. I also want to strengthen my ability to frame engineering problems as research problems by asking not only whether a system works, but why it works, when it fails, and how design choices affect reliability and usability.

In graduate school, I hope to contribute as someone who brings both initiative and range. I am comfortable learning new tools independently, but I also understand that real progress in research requires structure, critique, and rigor. I want to join a lab where I can help build prototypes, run experiments, analyze performance, and contribute to publications while developing into a stronger engineer and researcher. Over time, I hope to use that training to build technologies that can reduce labor burdens and widen opportunity in communities like mine.

I am applying to a research focused MS program in Mechanical Engineering because I already know the kind of problems I want to work on, and I have already begun preparing myself for them through projects, experimentation, and research. What I need now is the graduate training that will help me move from promising systems and strong motivation to deeper technical judgment and meaningful research contributions.

For more information on my projects, visit my portfolio ([portfolio Link](#)).